

Desca Affajry Rais

Mathematics Undergraduate · Researcher · Educator

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PROFESSIONAL SUMMARY

I am a final-year Mathematics undergraduate at Universitas Diponegoro with a 3.92/4.00 GPA. My primary interests are pure mathematics, especially real and complex analysis, graph theory, abstract algebra, topology, and combinatorics. I also have experience in university mathematics teaching, olympiad coaching, academic-content development, data analysis, and computational projects.

EDUCATION

Sep 2023 — Present

ONGOING

Universitas Diponegoro

B.Sc. in Mathematics · GPA 3.92 / 4.00

Chair, UKM KOMIPA UNDIP (2025); teaching and tutorial experience in Real Analysis II; active in the Mathematics Student Association; ON-MIPA delegate and mathematics competition awardee.

Jun 2020 — Mar 2023

COMPLETED

SMA Negeri 1 Sulang

Mathematics and Natural Sciences · Final Grade 86.55 / 100.00

Earned regional and national mathematics awards, represented the school in OSN Informatics, and provided peer mathematics mentoring.

2026

DRAFT

Distance Spectra of Corona Graphs of Complete Graphs and Paths via Invariant Subspaces

[Desca Affajry Rais](#) · [Suryoto, S.Si.](#), [M.Si.](#)

Studies the distance spectrum of $K_n \odot P_m$ using block matrices, Kronecker products, and invariant subspaces.

2026

DRAFT

Connectivity of Int-Graphic Topology on Directed Graphs

[Desca Affajry Rais](#) · [YD Sumanto](#) · [Solikhin](#)

Characterizes connectivity of finite int-graphic Alexandroff spaces through minimal open neighborhoods and the out-neighborhood inclusion preorder.

2026

PRESENTED

Modular Total Edge Irregular Labeling of Corona Product of Cycle C_4 with Path and Null Graphs

[The 16th ISNPINSA](#) · [Oral Presentation](#)

Developed explicit labeling constructions and established exact modular total edge irregularity strength results.

2026

COMPLETED

Inequality Problem Solving Book

[Desca Affajry Rais](#) · [Drs. YD Sumanto, M.Si.](#) · [First Edition](#) · [174 pages](#)

Proof-oriented mathematics book covering inequality theory, rigorous proofs, worked examples, and exercises. The manuscript is complete; Indonesian Intellectual Property (HKI) registration is planned.

PROFESSIONAL EXPERIENCE

Aug 2026 — Present	ONGOING University Mathematics Tutor Tutorkuliah.id Tutoring in advanced undergraduate mathematics with emphasis on conceptual understanding and rigorous problem solving.
Jan 2026 — Present	ONGOING Mathematics Olympiad Coach KOMIPA UNDIP University-level training in Real Analysis, Complex Analysis, Linear Algebra, Abstract Algebra, and Combinatorics.
Apr 2026 — Jun 2026	COMPLETED Teaching Assistant Asosiasi Matematika Terapan UNDIP Real Analysis tutorials, proof-oriented problem solving, and academic consultation.
Dec 2025 — Mar 2026	COMPLETED Mathematics Item Writer SainsIn Authored quantitative-reasoning items, detailed solutions, instructional videos, and quality-control materials.
Nov 2025 — Jan 2026	COMPLETED Data Analyst Emergencyy Call Data cleaning, SQL querying, Excel reporting, KPI dashboards, visualization, and workflow documentation.
Nov 2025 — Dec 2025	COMPLETED Mathematics Item Writer SNBT.ID Authored high-difficulty Quantitative Knowledge and Mathematical Reasoning items with structured solutions.
May 2025	COMPLETED Project-Based Virtual Intern · Data Scientist ID/X Partners × Rakamin Academy Built a credit-risk prediction pipeline using Logistic Regression and Random Forest.
Mar 2025	COMPLETED Project-Based Virtual Intern · Big Data Analyst PT Kimia Farma Tbk × Rakamin Academy Built a BigQuery analytical table and a Looker Studio performance dashboard.

LEADERSHIP & ORGANIZATIONS

Jan 2025 — Dec 2025

COMPLETED

Chairman

Mathematics and Sciences Olympiad Club (KOMIPA UNDIP)

Led strategy, daily operations, olympiad mentoring, ON-MIPA delegation, and external representation.

Jan 2025 — Dec 2025

COMPLETED

Senior Staff, Education and Reasoning Department

Mathematics Student Association, Universitas Diponegoro

Delivered mathematics tutorials, educational content, and learning-material development.

Feb 2024 — Jan 2025

COMPLETED

Junior Staff, Education and Reasoning Department

Mathematics Student Association, Universitas Diponegoro

Coordinated Academic Delegation, mentoring for 100+ students, and received Best Junior Staff and Best Department Staff recognition.

Oct 2023 — Dec 2024

COMPLETED

Staff, Networking and Public Relations Department

KOMIPA UNDIP

Supported ON-MIPA selection, served as Mathematics Liaison Officer, and led UKM Roadshow 2024.

2020 — 2023

COMPLETED

Moderator, Mentor, and Board of Governor

Brainly

Progressed through multiple moderation and mentoring roles, including supervision of 29 apprentice moderators.

2020 — 2022

COMPLETED

Organizational Leadership

Pelajar Islam Indonesia (PII), Rembang

Served as Head of Organizational Development and later General Secretary of the Rembang Regional Board.

COMMITTEE & VOLUNTEER EXPERIENCE

Facilitator Division Coordinator / Koordinator Divisi Fasilitator

27th Undip's Mathematics Competition · Jan 2025 — Feb 2026

Curriculum Division Staff / Staf Divisi Kurikulum

LKMM-TD FSM 2025 · Jan — Oct 2025

Event Division Staff / Staf Divisi Acara

Introduction, Gathering, & Adaptation Mathematics 2025 · Mar — Oct 2025

Event Division Staff / Staf Divisi Acara

Industrial Visit Mathematics 2025 · Mar — Aug 2025

Equipment Division Staff / Staf Divisi Perlengkapan

Math Action for Social and Education 2025 · Mar — Jun 2025

Public Relations Division Staff / Staf Divisi Humas

Leadership Training FSM 2025 · Mar — Apr 2025

Event Division Coordinator / Koordinator Divisi Acara

26th Undip's Mathematics Competition · Mar 2024 — Feb 2025

Chief Executive / Ketua Pelaksana

UKM Roadshow 2024 · Jun — Sep 2024

Volunteer Teacher / Pengajar Sukarela

Mathematics Clinic 2024 · Aug 2024

Equipment Division Coordinator / Koordinator Divisi Perlengkapan

ON-MIPA UNDIP Selection 2024 · Oct 2023 — Feb 2024

Mathematics Liaison Officer / Liaison Officer Matematika

ON-MIPA Coaching Center 2024 · Feb 2024

SELECTED PROJECTS

Feb 2026	COMPLETED Crowd Optimization and Smart Routing Strategy at Dufan Ancol Operations Research · Mathematical Modeling Applied M/M/s queueing, Little’s Law, greedy optimization, and knapsack concepts; proposed 14,000 daily visitors, a <45-minute waiting-time target, and 75% maximum utilization.
Feb 2026	ONGOING DMath Learning Mathematics Education · EdTech Ongoing mathematics learning platform for structured materials, guided practice, problem banks, olympiad preparation, and learning tools.
Feb 2026	COMPLETED Bike Sharing Analytics Python · EDA · Streamlit Identified weekday demand peaks, weekend casual-user behavior, and environmental drivers through end-to-end exploratory analysis.
Dec 2025	COMPLETED Financial Anomaly Detection System Scikit-Learn · K-Means · Random Forest Combined unsupervised pattern discovery with supervised transaction classification.
May 2025	COMPLETED Credit Risk Prediction Analysis SMOTE · Logistic Regression · Random Forest Analyzed 466,285 observations and reported Random Forest performance of 99.70% accuracy, 99.94% precision, and 99.46% recall.
Mar 2025	COMPLETED Kimia Farma Business Performance Analysis BigQuery · SQL · Looker Studio Built a centralized analytical table and multidimensional performance dashboard.
Mar–Apr 2025	COMPLETED Investment Portfolio Optimization Markowitz · Efficient Frontier · Sharpe Ratio Analyzed portfolio allocation across KLBF.JK, HEAL.JK, and KAEF.JK using risk–return optimization.

SCHOLARSHIPS & TRAINING

Nov 2024 — Present

ONGOING

BSI Scholarship Inspirasi Batch 4

Academic and leadership scholarship with coaching, mentoring, professional workshops, and social-impact activities.

Academic and leadership scholarship with coaching, mentoring, professional workshops, and social-impact activities.

Feb — Jul 2026

COMPLETED

Coding Camp 2026 — Data Scientist

Coding Camp powered by DBS Foundation

Coding Camp powered by DBS Foundation

Mar — Jun 2026

COMPLETED

Microsoft Elevate Training Center

Data Science with Microsoft Fabric · Generative AI with Microsoft Azure

Data Science with Microsoft Fabric · Generative AI with Microsoft Azure

Jan — Mar 2026

COMPLETED

IDCamp 2025 — Intermediate Data Scientist

Indosat Ooredoo Hutchison

Indosat Ooredoo Hutchison

Oct — Dec 2025

COMPLETED

IDCamp 2025 — Beginner Data Scientist

Indosat Ooredoo Hutchison

Indosat Ooredoo Hutchison

CERTIFICATIONS

UNDIP SEU English Proficiency Test EPT 65 · TOEFL Prediction equivalent 549 · 17 Sep 2026	TOEFL ITP Prediction Test 590 / 677
AI Engineer Bootcamp / Bootcamp AI Engineer	Foundations of Artificial Intelligence / Belajar Dasar AI
Foundations of SQL / Belajar Dasar SQL	Computer Networking Fundamentals / Fundamental Jaringan Komputer
Data Analyst — Basic Level ITBOX · Mar 2025	Python Programming Essentials: A Beginner's Guide Rakamin Academy · Mar 2025
Classical Cryptography for Beginners Cyber Academy Indonesia · Jul 2024	Introduction to Information Security Cyber Academy Indonesia · Jul 2024
Student Leadership and Management Training — FSM 2024 BEM FSM UNDIP · Apr 2024	Introduction to Data Science with Python DQLab · Jan 2024
Python Fundamentals for Data Science DQLab · Jan 2024	Basic Student Management Skills Training / LKMM Pra Dasar Sep 2023
Python Programming for Beginners CODEPOLITAN · Credential Z7R8TDD · Jul 2024 — Jul 2027	

SKILLS

Mathematics & Theory Pure Mathematics · Real Analysis · Complex Analysis · Graph Theory · Abstract Algebra · Topology · Combinatorics · Mathematical Modeling · Formal Proof Writing	Technical & Computing Python · SQL · LaTeX · Scikit-Learn · Looker Studio · Microsoft Excel · Microsoft Fabric · Microsoft Azure
Data & Analytics Data Analysis · Exploratory Data Analysis · Machine Learning · Data Visualization · Statistical Analysis · Data Engineering	Academic & Teaching Research · Academic Writing · Mathematics Olympiad · Teaching · Curriculum Development · Assessment Design · Problem Solving

AWARDS & ACHIEVEMENTS

Silver Medalist / Medali Perak · Isaac Newton Competition 2026 · OLIMNUS × UIN Sunan Kalijaga

Jul 2026

Silver Medalist / Medali Perak · Future Science Olympiad · POSI

Jun 2026

Bronze Medalist / Medali Perunggu · Mathematical Analysis and Geometry Day XVII · Institut Teknologi Bandung

May 2026

Gold Medalist / Medali Emas · Cendekia Natural Science Competition

Apr 2026

1st Place / Juara 1 · ON-MIPA UNDIP Selection 2026

Apr 2026

1st Place / Juara 1 · Indonesia Science Competition (KSI)

Mar 2026

Gold Award · KTOM March 2026

Mar 2026

1st Place / Juara 1 · Indonesia Science Competition (InSC)

Feb 2026

4th Place · Honorable Mention 1 · MATICS Mathematics Olympiad · Hasanuddin University

Feb 2026

Silver Award · KTOM February 2026

Feb 2026

Bronze Medalist / Medali Perunggu · POSI Science Competition

Jan 2026

2nd Place / Juara 2 · Indonesian Youth Science Olympiad #6

Jan 2026

2nd Place / Juara 2 · National Science League #2

Jan 2026

Silver Medalist / Medali Perak · Einstein Science Competition 2026

Jan 2026

Gold Award · KTOM December 2025

Dec 2025

Special Honour Participant · International Youth Math Challenge

Dec 2025

1st Place / Juara 1 · LOSEMA X

Nov 2025

1st Place / Juara 1 · Olimnus Science 2025

Nov 2025

Silver Medalist / Medali Perak · Isaac Newton Competition 2025

Jul 2025

3rd Place / Juara 3 · ASOE 4.0

May 2025

4th Place · ON-MIPA UNDIP Selection 2025

Mar 2025

Finalist / Finalis · MATHLICIOUS 2022

Nov 2022

1st Place / Juara 1 · ALHAMBRA Mathematics Competition

Feb 2022

Bronze Medalist / Medali Perunggu · Insight Arjuna Competition

Oct 2021

Gold Medalist / Medali Emas · Dirgantara Science Competition

Oct 2021

Silver Medalist / Medali Perak · Indonesian Youth Science Competition

Oct 2021

Bronze Medalist / Medali Perunggu · Bimage Competition Season 2

Sep 2021

Silver Medalist / Medali Perak · POSI Science Competition

Sep 2021

